

The Power of Penalties: Negativity-Aware Incentives for High-Quality Crowdsourced Data Labeling

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Abstract

High-quality data labeling is essential for training robust machine learning models; however, existing methods often ignore fraud or assume non-negative worker utility, failing to penalize harmful contributions without discouraging participation. To address this, we propose the Negativity-Aware Incentive (NAI) mechanism which introduces two novel components. First, the Ability-Result Characteristic Function (AR-CF) adapts and extends Shapley value theory through signed valuation to explicitly capture both positive and negative contributions, by combining workers' abilities with real-time task results to define contribution values. Second, a dynamic stake pool mechanism employs pre-commitment economics with adaptive dual-control parameters to balance fairness and operational efficiency. Through extensive experiments on multimodal datasets (images, text, audio, video), NAI outperforms state-of-the-art baselines: it improves video labeling accuracy by 16.6%, and reduces fraudulent behaviors by 33.9%. Furthermore, our deployment on the NetEase Youling crowdsourcing platform, serving 430,000 registered workers with 80,000 monthly active workers, validates NAI's real-world viability. Real-time A/B testing shows a 59.6% improvement in labeling quality for beginner tasks and a consistent reduction in fraud rates (14.8%-33.9%) across difficulty levels. This work establishes a paradigm shift in crowdsourcing system design, demonstrating that explicit negative modeling can enhance data quality, optimize costs, and foster participation at scale.

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CCS Concepts

• Information systems → Crowdsourcing; • Theory of computation → Algorithmic game theory and mechanism design.

Keywords

Crowdsourcing, Stake Pool, Mechanism design, Repeated game

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1 Introduction

Crowdsourcing has emerged as a pivotal technique for harnessing collective intelligence in data annotation tasks, enabling efficient processing of large-scale datasets for applications ranging from computer vision (e.g., image segmentation) [23] to natural language processing (NLP, e.g., sentiment analysis) [29]. Through platforms such as Amazon Mechanical Turk [1], requesters assign micro-tasks to global workers, who collectively label data, with the platform aggregating responses and managing compensation [19]. While this paradigm demonstrates remarkable scalability, its sustainability depends on a key challenge: how to equitably assess worker contributions and allocate rewards to ensure high-quality outcomes while maintaining ecosystem health [25].

Existing reward allocation mechanisms in crowdsourcing systems mainly follow two approaches. The first category adopts simple strategies such as equal distribution (i.e., fixed-rate payments) [5, 17], which inherently ignore differences in expertise and output quality among different workers. The second category attempts to address this limitation through performance-based compensation,

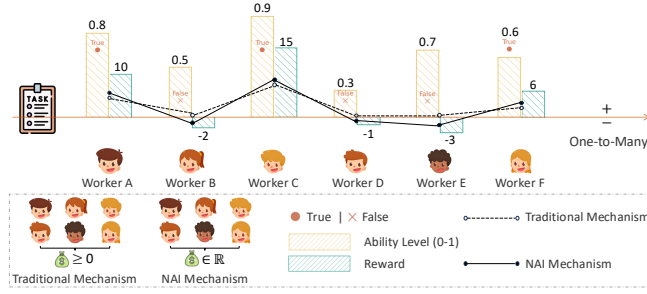


Figure 1: Illustration of NAI in crowdsourced data labeling.

evaluating workers based on historical accuracy [12], task-specific proficiency [6, 20], or temporal efficiency metrics [27]. Within Web economics and human computation, researchers have persistently highlighted systemic gaps in reporting equitable treatment of workers and compensation policy transparency [13, 14, 25]. Even the more sophisticated performance-based mechanisms still fail to account for two critical behavioral dimensions: 1) the dynamics of worker reliability, where previous trustworthy contributors may behave maliciously in new contexts [7]; 2) negative utilities arising from harmful behaviors that degrade collective outcomes [3]. Our analysis of Video Selection task instances [21] on the NetEase Youling¹ crowdsourcing platform reveals three systematic flaws in conventional approaches. First, fixed-rate schemes reward low-effort and high-quality workers equally, leading to unfair compensation. Second, performance-based models fail to capture behavioral volatility, as 22.7% of high-reputation workers demonstrated accuracy drops exceeding 25% in subsequent tasks. Most critically, both paradigms operate under an *individual rationality* assumption that presumes non-negative worker utilities, thereby ignoring two prevalent forms of negative contributions: (a) *passive negligence* (18.1% of workers showing less than 50% accuracy through random labeling) and (b) *active manipulation* (engaging in answer replication or adversarial patterns). These negative behaviors reduce the overall task quality by 42–48% compared to expert-only annotations, yet existing payment systems still reward malicious participants.

To address these critical gaps, our work advances a negativity-aware incentive mechanism guided by two fundamental research questions: (1) *How can worker contributions be dynamically quantified through a valuation framework that accounts for both positive and negative impacts?* Existing Shapley value-based methods assume non-negative marginal contributions [9, 24], making them ineffective for workers whose behavior alternates between helpful and harmful. Current systems fail to implement bidirectional evaluation, where correct answers should receive positive attribution and incorrect answers should be held negatively accountable. (2) *How to calibrate incentive allocation that fairly rewards positive contributions and proportionally penalizes negative impacts?* A naive proportional reward-penalty system may create dual disincentives: excessive penalties may lead to workers turnover, whereas insufficient penalties may increase platform costs. This calls for a mechanism balancing worker engagement and quality assurance, a dual-objective optimization challenge yet to be solved.

Our proposed Negativity-Aware Incentive (NAI) mechanism introduces two key innovations. First, the ability-result characteristic

function (AR-CF) jointly models workers’ historical capabilities and task-specific performance through a signed valuation framework. Unlike traditional Shapley approaches, AR-CF generates polarized contribution scores, which can explain differences between value-adding and value-subtracting behaviors [16]. Second, the stake pool mechanism introduces pre-commitment economics through mandatory point deposits, which aligns worker incentives with platform quality goals. Our dynamic parameter adjustment strategy employs dual control variables (penalty factor α and reward coefficient β) to maintain the incentive equilibrium between the platform and workers, thereby adaptively balancing fairness and operational feasibility. Figure 1 illustrates the NAI mechanism’s reward/penalty distribution in a one-to-many labeling scenario. For workers providing correct answers (A, C, F), both traditional and NAI mechanisms reward high-ability workers proportionally. However, for workers who make mistakes (B, D, E), conventional approaches preserve individual rationality by avoiding penalties, while NAI introduces differentiated accountability based on bounded rationality principles. Specifically, workers with higher abilities are penalized proportionally for incorrect answers, thus realizing the paradigm of “greater ability, greater responsibility”.

Our work makes the following contributions:

- We present the first systematic framework to address the challenge of penalizing negative contributions in crowdsourcing systems. By both encouraging high-quality output and disincentivizing value-reducing behaviors, we establish the theoretical foundation for high-quality data collection.
- We propose the Negativity-Aware Incentive (NAI) mechanism with two novel technical components. The Ability-Result Characteristic Function (AR-CF) adapts and extends the classical Shapley value theory by introducing a signed valuation mechanism that quantifies both positive and negative contributions through polarized scoring. As a complement to AR-CF, the stake pool mechanism introduces pre-commitment economics in crowdsourcing contexts, which creates a risk-return symmetry that combines individual rationality with collective quality goals.
- We validate NAI’s efficacy through large-scale real-world deployments on the NetEase Youling¹ crowdsourcing platform—serving 430,000 registered workers with 80,000 monthly active workers (as of Sep. 2025). Experimental results demonstrate consistent performance gains across diverse scenarios. In ability-critical tasks with the stake pool, NAI reduces fraud by 33.9% and improves labeling accuracy by 16.6% compared to the state-of-the-art baseline. For beginner-level tasks without penalty mechanisms, NAI still achieves 59.6% higher labeling quality through its capability-aware reward calibration. In addition, multimodal dataset evaluations (covering video, audio, text, and image labeling) show consistently high incentive fairness and substantial platform cost savings (Table 1), demonstrating the versatility of the framework across a variety of data modalities and task complexities.

2 Related Work

Game-Theoretic Foundations of Reward Allocation. Research on allocating rewards among workers traced back to cooperative game theory [2], where the Shapley value was paired with value-of-information models assuming non-negative marginal contributions

¹<https://zb.163.com/mark/task>

[24]. Recent work [9, 11] used the Shapley value to identify negatively contributing workers but stopped at evaluation, without informing allocation. Prior work [14] on worker compensation identified negative reinforcement/penalties as one structure, under which workers could lose a small portion of compensation for poor performance or inactivity. In such schemes, the worst case was zero pay, so workers who behaved fraudulently or manipulatively faced no real loss. In contrast, our mechanism provides an integrated evaluation-allocation framework that maps contribution estimates to payments and explicitly allows negative utility via a tunable penalty, creating real downside risk for harmful contributions.

Reward Allocation in Crowdsourcing Practice. Approaches to reward allocation in crowdsourcing were commonly categorized into fixed-rate payments [5, 17] and performance-based payments [6, 12, 27]. Fixed-rate payments compensated all workers equally, whereas performance-based payments tied compensation to historical proficiency [12], task-specific accuracy [6, 20], or time-based performance indicators [27], thus rewarding higher-ability workers more. However, these methods typically assumed non-negative worker utility and overlooked penalties for malicious or low-quality behavior; meanwhile, studies [14, 25] underscored the need for transparency and fairness in reward allocation to protect workers' rights. Our allocation rule departs from this assumption by incorporating negative utility directly into payment, yielding a mechanism that jointly accounts for positive and negative contributions, penalizing low-quality or adversarial behavior while rewarding high-quality work, thereby promoting fairness.

Incentive Mechanisms Beyond Allocation. Beyond reward allocation, incentive mechanisms in crowdsourcing largely focused on individual rationality (IR) by eliciting truthful reports of costs or effort [17, 22]. Prior studies examined and advocated fair compensation for crowd workers [5, 14]. Yet three issues remained: (i) IR typically enforced non-negative (expected) utility, (ii) requiring workers to report private costs/capacities complicated implementation and hindered deployment on commercial platforms, and (iii) prevailing payment rules offered little downside risk for fraudulent or manipulative workers. Our mechanism avoids eliciting private cost/capacity reports and introduces principled downside via negative utility, thereby improving deployability while aligning incentives against harmful behavior.

3 Problem Descriptions & Objectives

3.1 Problem Formulation

Consider a crowdsourced data labeling task involving three entities: a crowdsourcing server, a requester, and N workers (indexed by $1, 2, \dots, N$). The requester posts a set of L tasks on the platform, where each task $j \in \{1, 2, \dots, L\}$ requires a label $y_j \in \mathcal{A}$ (with $\mathcal{A}$ denoting the set of all possible answer options) and has an associated price p_j . Tasks may have varying difficulty levels. Each worker i possesses a latent skill θ_i that is normalized (via Item Response Theory (IRT) [18] and standardization techniques [10]²) to an ability $q_i \in (0, 1)$. Specifically, the probability that worker i correctly answers task j is modeled as $P(a_{ij} = y_j^* | \theta_i) = \frac{1}{1 + e^{-(\theta_i - b_j)}}$, where b_j denotes the difficulty parameter for task j . Each worker

i submits an answer $a_{ij} \in \mathcal{A}$ to the server. For task j , let $z_{ij} := \mathbb{I}(a_{ij} = y_j^*) \in \{0, 1\}$ denote whether worker i 's label is correct. The platform's objective is to maximize the labeling accuracy,

$$Q = \frac{1}{L} \sum_{j=1}^L \mathbb{I}(\hat{y}_j = y_j^*), \quad (1)$$

where $\hat{y}_j$ is the inferred label (obtained via truth inference algorithms [4, 28]) and y_j^* is the true label. Higher accuracy reduces task reissuance, lowers overall costs, and yields higher-quality labels. We aim to develop a reward-penalty system that incentivizes superior labeling. Our focus is twofold:

- (1) *Worker Contribution Evaluation.* We evaluate both positive and negative worker impacts using a *characteristic function*. An initial Correctness-Weighted Characteristic Function (CW-CF) was found inadequate for modeling negative effects (details in Appendix A). To address this, we formally define negative and positive contributions from both platform and coalition perspectives (see Appendix A) and later propose a more suitable Ability-Result Characteristic Function (AR-CF) in Section 4.1. Using the Shapley value framework [16], each worker's task-level contribution is computed as

$$\phi_{ij}(v) = \frac{1}{n} \sum_{S \subseteq N \setminus \{i\}} \binom{n-1}{|S|}^{-1} (v(S \cup \{i\}) - v(S)), \quad (2)$$

where positive Shapley values (ϕ_{ij}^+) indicate positive contributions, and negative values (ϕ_{ij}^-) indicate negative contributions.

- (2) *Reward/Penalty Distribution.* Given the positive and negative Shapley values, and taking into account factors such as worker participation and potential cost savings for the platform, the system distributes rewards or penalties to each worker.

We also account for potential *strategic behaviors* in data labeling, such as worker fraud and collusion, formally defined in Appendix A. Section 4 presents the complete mechanism, covering contribution evaluation, reward-penalty allocation, and anti-collusion measures.

3.2 Design Objectives

3.2.1 Fair Contribution Evaluation. Based on our platform practices and the principle of diminishing marginal contributions in cooperative games [15], the characteristic function v used for contribution evaluation should satisfy:

- **Negativity:** $v(S)$ can take negative for any coalition $S \subseteq N$.
- **Marginal Negativity:** A worker's marginal contribution may be negative, i.e., $v(S \cup \{i\}) - v(S) < 0$.
- **Marginal Monotonicity:** For two workers i, i' with $q_i \geq q_{i'}$: if both are correct, $v(S \cup \{i\}) - v(S) \geq v(S \cup \{i'\}) - v(S)$; if both are incorrect, the inequality reverses.

These properties ensure that high-ability workers gain large positive contributions for correct answers and strong penalties for incorrect ones, given their greater impact on aggregated confidence. Low-ability workers, by contrast, receive minimal or negative contributions.

3.2.2 Equitable Reward Allocation. Our reward mechanism must provide fair economic feedback to workers and maintain platform efficiency. In particular, it should satisfy:

²[10] standardizes and transforms IRT parameters for stability and consistency.

- **Redistributive Incentive:** A fraction of the penalty collected from workers with negative contributions is redistributed among workers with positive contributions (e.g., in proportion to their abilities), while the remaining is retained by the platform.
- **Sloppiness:** Workers may incur negative utility, i.e., $U_i \in \mathbb{R}$ with $U_i < 0$ possible.
- **Truthfulness/Incentive Compatibility:** A mechanism M is truthful if, for every worker i , truthfully reporting their private information yields utility at least as high as any misreport. Formally, for all i , alternative reports τ'_i , and other workers' reports τ_{-i} : $U_i(\tau_i, M(\tau_i, \tau_{-i})) \geq U_i(\tau'_i, M(\tau'_i, \tau_{-i}))$.
- **Computational Efficiency:** The mechanism is executable in polynomial time.

4 NAI Mechanism

This section formalizes the Negativity-Aware Incentive (NAI) mechanism and presents its necessary prerequisite, truth inference, as illustrated in Figure 2. We first define the Ability-Result Characteristic Function (AR-CF), then describe the dynamic Stake Pool, and finally outline the reward and penalty distribution scheme.

4.1 Ability-Result Characteristic Function

The AR-CF is defined as

$$v(S) = \frac{1}{|S|} \sum_{i \in S} (z_i q_i - (1 - z_i) q_i), \quad (3)$$

which is intuitively grounded in two perspectives:

- (1) *Platform Perspective.* The normalized ability q_i serves as a proxy for worker i 's expected accuracy. Thus, $z_i q_i$ represents credit for a correct answer, while $-(1 - z_i) q_i$ represents a penalty for an incorrect answer. In aggregate, $v(S)$ approximates the overall quality (or confidence) of the coalition's responses; a drop in $v(S)$ upon adding a worker indicates that the worker degrades the overall confidence.
- (2) *Coalition Perspective.* Within a group, if a worker i has an ability q_i lower than the average ability of the coalition, then i 's inclusion dilutes the overall performance. That is, if

$$q_i < \bar{q}_S = \frac{1}{|S|} \sum_{k \in S} q_k,$$

then the marginal effect of adding i is negative.

We analyze the practical implications of critical values of $v(S)$:

- $v(S) \rightarrow -1$: Occurs when all workers have abilities approaching 1 but answer incorrectly.
- $v(S) = 0$: Occurs when the contributions of correct and incorrect answers balance.
- $v(S) \rightarrow 1$: Occurs when all workers have abilities approaching 1 and answer correctly.

Moreover, due to diminishing marginal returns in worker cooperation, a larger coalition is not necessarily better. Instead, higher ability among workers is more beneficial. Higher abilities increase the likelihood of correct answers, leading to larger positive contributions and enhancing the coalition's overall performance.

The proofs for negativity, marginal negativity, and marginal monotonicity of AR-CF are straightforward and are omitted for brevity.

THEOREM 4.1. *AR-CF satisfies negativity, marginal negativity, and marginal monotonicity.*

4.2 Dynamic Stake Pool

The *Stake Pool* is a key component of our NAI mechanism, handling:

- (1) *Staking Requirement:* Before participating, workers must pledge points as collateral to cover potential penalties.
- (2) *Determination of α and β :* Dynamically setting the penalty factor α and the reward coefficient β . Here, α scales the task price to determine the total penalty, while β modulates the fraction of the penalty redistributed as a bonus.
- (3) *Reward and Penalty Distribution:* After evaluating worker contributions via AR-CF and computing Shapley values, the Stake Pool partitions these values into positive and negative components, normalizes them, and then distributes rewards and penalties accordingly.

4.2.1 Staking Requirement. Assume that each task j has price p_j , with prices distributed according to a density $g(p_j)$ over $[p_{\min}, p_{\max}]$. Let $\bar{p} = E[p_j]$ denote the expected task price, and let $\bar{m}$ be the average number of tasks per worker. In the worst-case scenario (if a worker answers all $\bar{m}$ tasks incorrectly), the total penalty incurred is approximately

$$U_i^{\max} = \bar{m} \cdot \alpha \bar{p}.$$

Thus, to ensure that no worker's loss exceeds their staked amount, the fixed stake S must satisfy

$$S \geq \bar{m} \cdot \alpha \bar{p}. \quad (4)$$

Alternatively, in the dynamic staking mode, we set the stake for worker i as

$$S_i = \bar{m} \cdot \alpha \bar{p} (1 - q_i), \quad (5)$$

so that higher-ability workers, who are less likely to incur penalties, pledge fewer points.

Determination of α and β . To determine the optimal values of the penalty factor α and the reward coefficient β , we first introduce a critical variable—the *threshold ability level* q^* . This threshold denotes the minimum ability a worker must have for their expected net reward per task to be nonnegative. From the platform's perspective, workers with abilities above q^* are more likely to produce accurate labels, thereby exerting a greater positive influence on the confidence of aggregated results. Consequently, q^* serves as a natural partitioning point in our model for determining α and β . Accordingly, this part is twofold: (a) *Threshold Ability Level for Non-Negative Net Reward* – We derive q^* by setting the expected net reward per task to zero. (b) *Dynamic Parameter Adjustment Strategy* – We then propose a strategy for dynamically adjusting α and β in practice to determine their optimal values.

4.2.2 Threshold Ability Level for Non-Negative Net Reward. We derive a closed-form threshold ability q^* such that a worker's expected net reward per task is nonnegative (derivation in Appendix B).

$$q^* = \frac{\alpha \mu}{\alpha \mu + (\beta \alpha + 1)(1 - \mu)}. \quad (6)$$

When $q > q^*$, expected bonuses dominate penalties; otherwise penalties dominate.

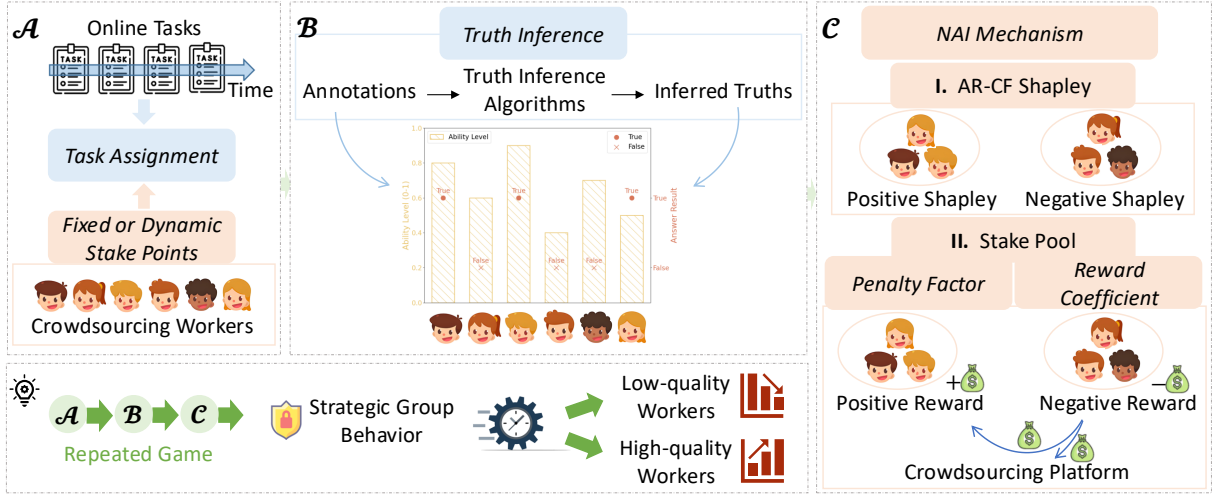


Figure 2: High-level overview of the NAI-powered crowdsourcing platform.

4.2.3 Dynamic Parameter Adjustment Strategy. Based on the threshold analysis, we dynamically optimize α and β to maximize the platform's utility under quality, participation, profitability, and risk constraints (see Appendix C).

4.2.4 Reward and Penalty Distribution. After the optimal parameters α and β are determined, the Stake Pool implements the reward and penalty distribution among workers for each task. For task j , let N_j denote the set of participating workers. After computing each worker's contribution using AR-CF, we decompose the Shapley value ϕ_{ij} into its positive and negative parts, denoted by ϕ_{ij}^+ and ϕ_{ij}^- , respectively.

We then allocate to worker i for task j a reward

$$r_{ij} = (\beta \alpha p_j + p_j) \cdot \frac{\phi_{ij}^+}{\sum_{l \in N_j} \phi_{lj}^+}, \quad (7)$$

or a penalty

$$d_{ij} = \alpha p_j \cdot \frac{\phi_{ij}^-}{\sum_{l \in N_j} \phi_{lj}^-}. \quad (8)$$

Over all tasks, let $T_i^+ \subseteq \{1, 2, \dots, L\}$ and $T_i^- \subseteq \{1, 2, \dots, L\}$ denote the sets of tasks where worker i receives positive and negative Shapley values, respectively. The worker's utility is defined as:

$$U_i = R_i - D_i = \sum_{j \in T_i^+} r_{ij} - \sum_{j \in T_i^-} d_{ij}, \quad (9)$$

where R_i and D_i represent the total reward and total penalty of worker i , respectively. If $U_i < 0$, the corresponding amount is deducted from their staked points (as described in Section 4.2.1); otherwise, the worker retains their stake and receives the surplus.

One key innovation of our mechanism is the *Redistributive Incentive Property*: the platform retains a portion $(1 - \beta)\alpha p_j$ of the total penalty, while redistributing the remainder to workers with positive contributions. This allows the platform to reduce costs by:

$$C = \sum_{j \in T_i^-} (1 - \beta)\alpha p_j. \quad (10)$$

4.3 Formulation of the NAI Mechanism

The NAI mechanism couples AR-CF with a dynamically optimized *Stake Pool*. It operates over three task tiers: *Beginner* (no history/points), *Ability* (with history/points), and *Staked/Challenging* (Ability + stake). The workflow is summarized in Algorithm 1.

NAI's incentive guarantees rest on two standard premises from truth inference theory [4, 28]:

ASSUMPTION 1 (CONDITIONAL INDEPENDENCE). Worker responses are conditionally independent given the true label y_j^* , i.e.,

$$P(\{a_{ij}\}_{i=1}^N | y_j^*) = \prod_{i=1}^N P(a_{ij} | y_j^*).$$

This enables reliable estimation of worker abilities q_i through expectation-maximization methods [4].

ASSUMPTION 2 (MAJORITY CONSENSUS). For any task j , the majority of workers provide correct answers:

$$\sum_{i=1}^N \mathbb{I}(a_{ij} = y_j^*) > \sum_{i=1}^N \mathbb{I}(a_{ij} \neq y_j^*).$$

This holds when worker abilities satisfy $q_i > 0.5$ on average [28].

We demonstrate that NAI Mechanism is redistributive incentive, sloppy, truthful, and computationally efficient. Detailed proofs are provided in Appendix F.

THEOREM 4.2. NAI Mechanism is redistributive incentive, sloppy, truthful/incentive-compatible, and computationally efficient.

4.4 Truth Inference & Information Elicitation

For each task j , labels are inferred in real time using standard truth inference / information elicitation methods [4, 8, 26, 28].

4.4.1 Robust Truth Inference Framework. To ensure reliable labeling under adversarial conditions, we employ a layered defense:

THEOREM 4.3 (PRIOR ABILITY PROFILING MITIGATES COLLUSION). Let T be the number of historical tasks used to evaluate each worker's accuracy. Let $\hat{q}_i = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(a_{it} = y_t^*)$ denote the historical accuracy

ALGORITHM 1: NAI-Driven Crowdsourced Data Labeling

Input: Tasks $\mathcal{T} = \{1, \dots, L\}$ with price p_j ; workers $N = \{1, \dots, n\}$ with ability q_i ; responses $\{a_{ij}\}$.
Output: utility for each worker i and platform cost savings.
// Phase 1: Assignment & Staking
if worker $i \in \{\text{Beginner}, \text{Ability}\}$ **then**
| Disable stake pool for i ;
end
else if worker $i \in \{\text{Staked}, \text{Challenging}\}$ **then**
| Set required stake points via Eq. (4) or (5);
end
// Phase 2: Truth Inference
foreach task $j \in \mathcal{T}$ **do**
| Infer label of j from $\{a_{ij}\}_{i \in N_j}$ (e.g., Dawid–Skene [4]);
end
// Phase 3: Ability Update
foreach worker $i \in N$ **do**
| Update q_i based on the current task performance;
end
// Phase 4: Stake Pool Tuning
Solve optimization (Sec. 4.2) for α and β ;
// Phase 5: Contribution Evaluation
foreach task $j \in \mathcal{T}$ **do**
| Compute coalition AR-CF and individual Shapley ϕ_{ij} ;
end
// Phase 6: Reward/Penalty & Utility
foreach task $j \in \mathcal{T}$ **do**
| **foreach** worker $i \in N_j$ **do**
| | Assign reward by Eq. (7) and penalty by Eq. (8);
| **end**
end
foreach worker $i \in N$ **do**
| Compute utility over all tasks via Eq. (9);
| **if** utility < 0 **then**
| | Deduct $|\text{utility}|$ from worker i 's staked points;
end
// Phase 7: Platform Savings
Compute platform cost savings via Eq. (10);
return $\{\text{utility}_i\}_{i \in N}$ and platform cost savings;

of worker $i \in N$. Define the trusted worker set $\mathcal{W}_{\text{trusted}} = \{i \in N \mid \hat{q}_i \geq \tau\}$, $\tau \in (0, 1)$, and assign weights $w_i = \begin{cases} 3q_i, & i \in \mathcal{W}_{\text{trusted}}, \\ q_i, & \text{otherwise.} \end{cases}$ If the average weight of trusted workers is at least 3τ and that of colluders is below τ , then

$$\frac{W_{\text{trusted}}}{W_{\text{colluder}}} \geq 3 \cdot \frac{|\mathcal{W}_{\text{trusted}}|}{|\mathcal{W}_{\text{colluder}}|},$$

thus suppressing colluder influence.

THEOREM 4.4 (GOLD STANDARD QUESTIONS MITIGATE COLLUSION). Let B be a batch of tasks and let $B_{\text{gold}} \subset B$ be a subset of $\kappa|B|$ gold standard tasks (with $0 < \kappa \leq 1$) whose correct answers y_j^{gold} are

known. For each worker i , define the credibility score as

$$c_i = 1 - \frac{1}{\kappa|B|} \sum_{j \in B_{\text{gold}}} \mathbb{I}(a_{ij} \neq y_j^{\text{gold}}).$$

If a colluding worker's correct-answer probability $s_i < 0.7$, Hoeffding's inequality implies that $\Pr(c_i \geq 0.7)$ decays exponentially in $\kappa|B|$. Setting threshold $c_{\min} = 0.7$ and excluding workers with $c_i < c_{\min}$ effectively removes colluders and low-quality participants.

THEOREM 4.5 (ANCHOR WORKERS MITIGATE COLLUSION). Let $\mathcal{W}_{\text{anchor}} \subset N$ be a certified (anchor) worker pool with $|\mathcal{W}_{\text{anchor}}| = \eta N$, $\eta \in [0.05, 0.1]$. For answer option $k \in \mathcal{A}$,

$$n_k = \sum_{i \in \mathcal{W}_{\text{anchor}}} \mathbb{I}(a_{ij} = k).$$

If the true label is k^* and $n_{k^*} \geq \sum_{i \in \mathcal{W}_{\text{anchor}}} \mathbb{I}(a_{ij} = k') + 1$ for all $k' \neq k^*$, then the consensus prior

$$P(y_j^* = k) \propto \exp(\lambda n_k), \quad \forall k \in \mathcal{A},$$

satisfies

$$\frac{P(y_j^* = k^*)}{P(y_j^* = k')} = \exp(\lambda(n_{k^*} - n_{k'})) \geq \exp(\lambda).$$

This exponential bias in favor of the true label k^* mitigates collusion.

Detailed proofs are provided in Appendices F–F.

4.4.2 Operational Enforcement. Penalties escalate for colluders:

- First offense: $D_i \leftarrow 2D_i$ (double penalty).
- Second offense: Account suspension for suspensionDays = 30.
- Third offense: Permanent ban and forfeiture of staked points.

Our layered defense—weighted inference, anchor workers, and penalty enforcement—achieves reliable label estimation even under manipulation. In deployment (Section 5.2), NAI reduces collusion (measured by QWAB, the quantity of workers with ability below 0.5) by 83% compared to baseline mechanisms.

4.5 NAI as a Repeated Game

Over repeated rounds, NAI retains high-quality workers and filters out low-quality ones (see Appendix E for details).

5 Evaluation and Application

In this section, we present extensive experimental results on real-world datasets and describe the deployment of our NAI mechanism on a commercial crowdsourcing platform. Our evaluation and application aims to validate the following research questions:

- (1) RQ1: To what extent does AR-CF reflect worker contributions?
- (2) RQ2: What is NAI's impact on quality and low-quality behavior?
- (3) RQ3: How do negative rewards trade off participation and fraud?

5.1 Experimental Evaluation

5.1.1 Experimental Setup. We utilize real annotation data from the NetEase Youling¹ crowdsourcing platform covering four task types: image classification, text selection, audio classification, and video selection [21]. For each modality, one task set is selected (with 1,000 samples for image classification, 18,398 for text selection, 4,789 for audio classification, and 500 for video selection). Each task corresponds to a single question with exactly one correct answer.

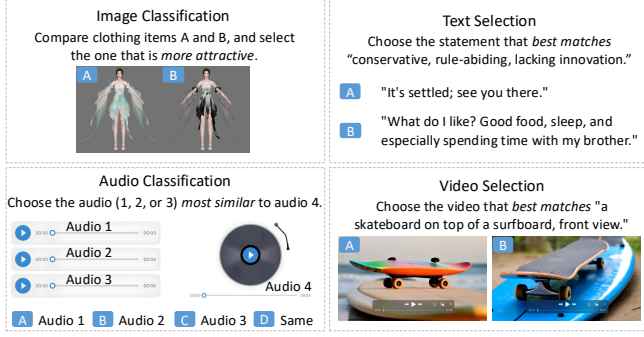


Figure 3: Examples of Annotation Tasks.

Real examples of questions, including multiple-choice classification questions, are shown in Figure 3. The number of workers per question ranges from 4 to 31, based on a histogram from 2,200 randomly sampled tasks (see Figure 4). The image classification task set has no worker-ability threshold, whereas the other three sets do.

Baselines. To benchmark our contributions, we compare our proposed AR-CF and NAI mechanisms against three competitive baselines. Among them, AS Mechanism and PS Mechanism are considered state-of-the-art methods:

- *AS Mechanism*: A fixed-rate payment scheme equally dividing the total reward among workers who answer correctly [5].
- *PS Mechanism*: A performance-based payment scheme allocating rewards proportionally to workers' estimated abilities [12, 27].
- *CW Mechanism*: A scheme based on the Correctness-Weighted Characteristic Function (CW-CF, see Section 3.1).

Performance Metrics. We assess our approaches using the following metrics:

- *Reasonableness Metric*: The Shapley value ϕ_i for each worker i is computed based on the chosen characteristic function, reflecting the average marginal contribution of each worker.
- *Fairness Metric*: The fairness of reward allocation is quantified by the Spearman rank correlation coefficient F between the per-worker average *net utility* vector $\bar{u} = [\bar{u}_1, \bar{u}_2, \dots, \bar{u}_n]$ and the ability vector $\mathbf{q} = [q_1, q_2, \dots, q_n]$, where $\bar{u}_i = \frac{1}{|T_i|} \sum_{j \in T_i} (r_{ij} - d_{ij})$ is worker i 's average net utility across all tasks T_i they participated in. A value of F close to 1 indicates that higher-ability workers receive proportionally higher net rewards.
- *Platform Cost Savings (C)*: The total cost savings achieved by the platform from the fraction of the penalty retained (i.e., $(1 - \beta)\alpha p_j$) over all tasks.

5.1.2 Evaluation of AR-CF. To address RQ1, we compare the Shapley values computed using our proposed AR-CF with those obtained using CW-CF across all four task types. For each worker i , we compute the average per-task Shapley value ϕ_i and plot it against their ability q_i . As illustrated in Figure 5, AR-CF assigns negative contributions to low-ability workers and provides higher contribution values for high-ability workers compared to CW-CF. In particular, in tasks with an enforced answer threshold, AR-CF effectively penalizes workers who, despite meeting the threshold, underperform, thereby yielding a more discriminative evaluation of worker contributions.

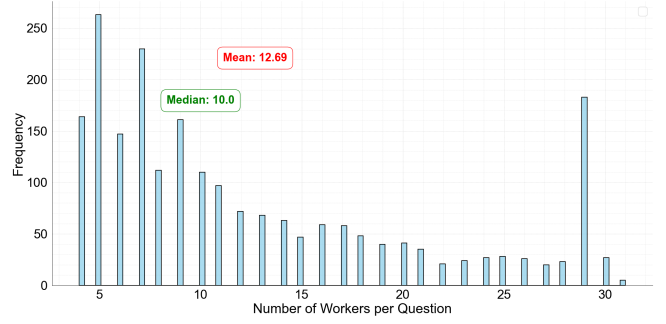


Figure 4: Distribution of Workers per Task.

Table 1: Fairness Metric F & Cost Savings C for Mechanisms.

Mechanism	Fairness Metric F				Cost Savings C
	Image	Text	Audio	Video	
NAI	0.94	0.98	0.97	0.99	6400.40
CW	0.95	0.95	0.96	0.97	6345.60
PS	0.98	0.98	0.94	0.98	0.00
AS	0.95	0.93	0.86	0.95	0.00

5.1.3 Evaluation of NAI Mechanism. To address RQ2 and RQ3, we compare reward allocation under NAI, CW, PS, and AS with $p = 1$, $\alpha = 0.5$, and $\beta = 0.2$. For each worker i , we compute the average *net utility* per task $\bar{u}_i$ under each mechanism and plot $\bar{u}_i$ against ability q_i (Figure 6). The results show that: (a) NAI assigns lower, and sometimes negative, rewards to low-ability workers, effectively penalizing poor performance; and (b) NAI rewards high-ability workers substantially more than the other mechanisms.

Table 1 further shows that: (a) all mechanisms achieve high fairness correlations ($F > 0.85$) since higher-ability workers are more likely to answer correctly and all mechanisms reward correctness; (b) AS (equal split among correct workers) yields the lowest F , especially for Audio (0.86), indicating that equal division fails to reflect ability differences; and (c) NAI and CW yield large platform cost savings C (over 6000 points across tasks) by retaining $(1 - \beta)\alpha p_j$ of the penalties, whereas PS and AS yield zero savings because they cannot incorporate penalties.

5.2 Application on a Commercial Platform

To validate NAI in practice and further answer RQ2–RQ3, we deployed it on the NetEase Youling¹ crowdsourcing platform and conducted online A/B tests on two task types: staked Ability Tasks and non-staked Beginner Tasks.

5.2.1 Online A/B Testing of Ability Tasks. We evaluated different reward allocation mechanisms on staked ability tasks. **Factor:** Reward allocation mechanism. **Variants:** (1) *Control (NAI Mechanism without staking)*: Ability tasks where negative rewards are disabled. (2) *Treatment A (CW Mechanism with staking)*: Staked ability tasks using the CW mechanism with dynamic stake points. (3) *Treatment B (NAI Mechanism with staking)*: Staked ability tasks using the NAI mechanism with dynamic stake points. **Environment:** For each variant, 500 video selection ability tasks were posted at 10 AM and 500 at 3 PM daily over one month. Each task required between 3 and 28 participants, with $p = 2$, $\alpha = 0.5$, and $\beta = 0.2$.

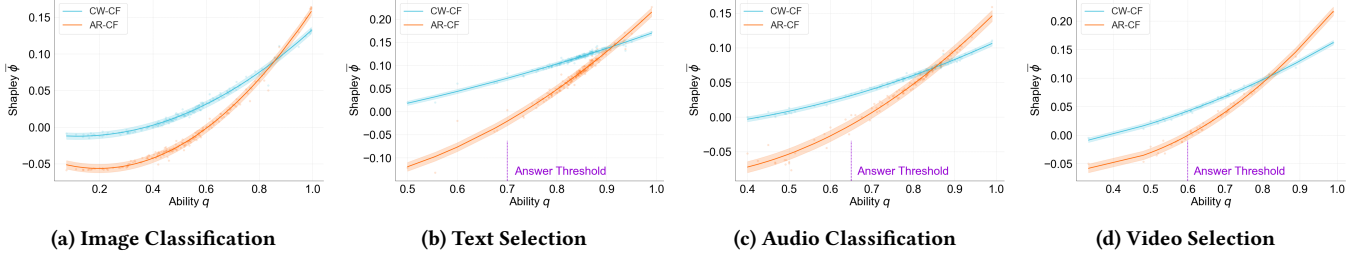


Figure 5: Comparison of AR-CF and CW-CF Shapley values across ability levels in four task types.

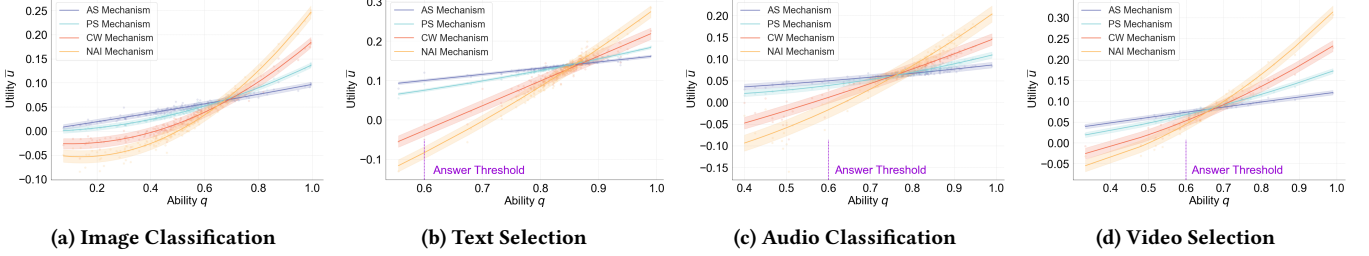


Figure 6: Comparison of NAI, CW, PS, and AS mechanisms across ability levels in four task types.

Metrics: The following metrics were observed: (a) *NCT*: Number of Completed Tasks (label confidence ≥ 0.99 via truth inference). (b) *QWAB*: Quantity of Workers with Ability Below 0.5. (c) *NWAB*: Number of Workers with Ability Between 0.5 and 0.6. (d) *AAW*: Average Ability of Workers.

Higher *NCT* reflects improved accuracy in the aggregated answers and, hence, better data labeling quality. The significant decreases in *QWAB* and *NWAB* indicate that fraudulent and manipulative behaviors have been curtailed, while the increase in *AAW* demonstrates that workers are effectively incentivized.

Table 2: Online A/B Testing of Ability Tasks on the NetEase Youling Crowdsourcing Platform.

Variant	Mechanism	NCT	QWAB	NWAB	AAW
Ctl.	NAI	17,878	357	292	0.68
Trt. A	CW (Stake)	18,413	236	121	0.69
Trt. B	NAI (Stake)	21,477	62	87	0.72

As shown in Table 2, from Sep. 1 to Sep. 30, 2025, online A/B testing shows that treatment B (NAI with staking) outperforms the other variants by increasing *NCT* by approximately 20%, boosting *AAW* by about 6%, and reducing *QWAB* and *NWAB* by 83% and 70%, respectively.

5.2.2 Online A/B Testing of Beginner Tasks. We evaluated different reward allocation mechanisms on beginner tasks. **Factor:** Reward allocation mechanism. **Variants:** (1) *Control (AS Mechanism)*: Workers receive 0 points for incorrect answers and share rewards equally for correct answers. (2) *Treatment A (PS Mechanism)*: Workers receive 0 points for incorrect answers and are rewarded in proportion to their relative abilities for correct answers. (3) *Treatment B (NAI Mechanism)*: Beginner tasks processed using our proposed mechanism. **Environment:** For each variant, 500 video selection beginner tasks were posted at 10 AM and 500 at 3 PM daily over one month.

Each task required between 3 and 28 participants, with $p = 1$. **Metrics:** The same metrics as for Ability Tasks (*NCT*, *QWAB*, *NWAB*, and *AAW*) were observed.

Table 3: Online A/B Testing of Beginner Tasks on the NetEase Youling Crowdsourcing Platform.

Variant	Mechanism	NCT	QWAB	NWAB	AAW
Ctl.	AS	1,477	791	765	0.55
Trt. A	PS	2,038	660	762	0.57
Trt. B	NAI	3,253	562	650	0.61

As shown in Table 3, from Jun. 1 to Jun. 30, 2025, online A/B testing shows that treatment B (NAI) outperforms the other variants by increasing *NCT* by approximately 1.20 times, boosting *AAW* by about 11%, and reducing *QWAB* by 29% and *NWAB* by 15%.

6 Conclusion

In this paper, we addressed fraud and manipulation in crowdsourced data labeling with NAI, combining AR-CF (a signed contribution function) with stake-based allocation. NAI rewards helpful work and penalizes harmful work while bounding worker liability and sustaining platform viability through adaptive stake levels. Experiments on real-world datasets and a commercial deployment validate its effectiveness. Future work will extend NAI to other marketplaces and integrate stronger real-time fraud detection.

Acknowledgments

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A Supplementary Definitions

Correctness-Weighted Characteristic Function (CW-CF).

$$v(S)_{\text{CW}} = \frac{\sum_{i \in S} q_i z_i}{\sum_{i \in S} q_i}, \quad (11)$$

$S \subseteq N$ is a worker coalition. CW-CF rewards correctness but ignores errors, failing to capture negative contributions, thus motivating the Ability–Result Characteristic Function (AR-CF) in Section 4.1.

Definition A.1 (Negative and Positive Contributions). For any coalition $S \subseteq N$, consider adding worker $i \notin S$. We say that worker i contributes negatively to S if either:

- (1) *(Platform Perspective)*: Including i reduces the confidence of the aggregated answer, i.e., $C_j(S \cup \{i\}) < C_j(S)$, where $C_j(S)$ is the overall confidence of the aggregated answer for task j given workers S .
- (2) *(Coalition Perspective)*: Worker i 's ability is below the average ability of S , i.e., $q_i < \bar{q}_S = \frac{1}{|S|} \sum_{k \in S} q_k$, so that i 's inclusion dilutes the overall quality of the coalition.

Otherwise, i contributes positively.

Definition A.2 (Worker Fraud and Manipulation). Let n_i denote the total number of tasks completed by worker i , and let

$$f_i = \frac{\max_{k \in \mathcal{A}} \{\text{number of times } i \text{ selects option } k\}}{n_i},$$

where $\mathcal{A}$ denotes the set of all possible answer options, and k is any element in $\mathcal{A}$. We say worker i engages in *fraud or manipulation* if at least one of the following conditions holds:

- **Rapid Response Behavior**: The ratio of tasks completed with response time below a threshold $t_{\min}$ is high: $\frac{n_i^{\text{rapid}}}{n_i} \geq \gamma_1$, where n_i^{rapid} is the number of tasks completed in less than $t_{\min}$ seconds and $\gamma_1 \in (0, 1)$ is a preset constant.
- **Answer Concentration**: The proportion f_i exceeds a threshold γ_2 (e.g., $\gamma_2 = 0.9$), indicating an excessively high frequency of the same answer.
- **Low Empirical Accuracy**: Worker i 's empirical accuracy is below threshold γ_3 (e.g., 0.5) over a sufficient number of tasks, despite a higher expected accuracy from historical performance.

Definition A.3 (Worker Collusion). Consider a coalition $S \subseteq N$ of workers. We say that S is colluding if its members systematically coordinate their responses to manipulate the aggregated outcome or increase their net rewards, deviating from their individually optimal (truthful) behavior. Formally, assume there exists a coordination function $\sigma: \{1, 2, \dots, L\} \rightarrow \mathcal{A}$ such that for every task j and each $i \in S$,

$$\mathbb{P}(a_{ij} = \sigma(j)) = 1.$$

If, for a non-negligible fraction of tasks, the coordinated response $\sigma(j)$ consistently differs from the true label y_j^* (or from what would

be independently selected based on q_i), then S is deemed to be colluding. Equivalently, if for some $k \in \mathcal{A}$ and many tasks,

$$\mathbb{P}\left(\bigcap_{i \in S} \{a_{ij} = k\}\right) \gg \prod_{i \in S} P_i(a_{ij} = k),$$

where $P_i(a_{ij} = k)$ is the probability that worker i independently selects k , the observed statistical dependence signals collusion.

B Derivation of the Threshold Ability q^*

Assume worker abilities q are distributed over $(0, 1)$ with density $f(q)$. For a worker with ability q , we assume the expected positive Shapley contribution per task is proportional to q and the expected negative contribution is proportional to $(1 - q)$, i.e.,

$$E[S^+(q)] = k^+ q E[p_j], \quad E[S^-(q)] = k^- (1 - q) E[p_j],$$

where k^+ and k^- are constants representing the maximum contributions per task.

The total expected positive Shapley value aggregated over all W workers is

$$S_{\text{total}}^+ = W \int_0^1 E[S^+(q)] f(q) dq = k^+ W E[p_j] \mu,$$

and the total expected negative Shapley value is

$$S_{\text{total}}^- = k^- W E[p_j] (1 - \mu),$$

where $\mu = \int_0^1 q f(q) dq$.

Thus, the proportions of the total Shapley values for a worker with ability q are:

$$\phi^+(q) = \frac{E[S^+(q)]}{S_{\text{total}}^+} = \frac{q}{W\mu}, \quad \phi^-(q) = \frac{E[S^-(q)]}{S_{\text{total}}^-} = \frac{1 - q}{W(1 - \mu)}.$$

Under our reward mechanism, the expected reward from positive contributions per task is:

$$E[r] = (\beta\alpha + 1)E[p_j] \phi^+(q) = (\beta\alpha + 1)E[p_j] \cdot \frac{q}{W\mu},$$

and the expected penalty from negative contributions is:

$$E[d] = \alpha E[p_j] \phi^-(q) = \alpha E[p_j] \cdot \frac{1 - q}{W(1 - \mu)}.$$

Hence the expected net reward per task for a worker of ability q is

$$E[D(q)] = E[r] - E[d] = \frac{E[p_j]}{W} \left(\frac{(\beta\alpha + 1)q}{\mu} - \frac{\alpha(1 - q)}{1 - \mu} \right). \quad (12)$$

Setting $E[D(q^*)] = 0$ yields

$$\frac{(\beta\alpha + 1)q^*}{\mu} = \frac{\alpha(1 - q^*)}{1 - \mu} \Rightarrow q^* = \frac{\alpha\mu}{\alpha\mu + (\beta\alpha + 1)(1 - \mu)},$$

which matches Eq. (6) in the main text.

C Optimization of α and β

We tune (α, β) by maximizing a weighted utility

$$U(\alpha, \beta) = w_1 Q(\alpha, \beta)^2 + w_2 L(\alpha, \beta)^2 + w_3 R(\alpha, \beta) + w_4 Q(\alpha, \beta)L(\alpha, \beta) - w_5 C_{\text{risk}}(\alpha, \beta) + w_6 \alpha, \quad (13)$$

where $w_k \in [0, 1]$ encode platform priorities; squares discourage low quality/participation and the interaction term rewards jointly high Q and L .

Components. Let $q^* = q^*(\alpha, \beta)$ be the ability threshold (Eq. (6)), f/F the pdf/cdf of abilities, $\bar{p} = E[p_j]$, and $m(q) = m_{\max} q$.

$$Q(\alpha, \beta) = \int_{q^*}^1 q f(q) dq, \quad L(\alpha, \beta) = \int_{q^*}^1 f(q) dq = 1 - F(q^*)$$

$$R(\alpha, \beta) = (1 - \beta) \alpha \bar{p} \int_0^{q^*} m(q) f(q) dq$$

Dynamic staking risk (expected shortfall under stakes $S_i \propto (1 - q)$):

$$C_{\text{risk}}(\alpha, \beta) = \bar{m} \alpha \bar{p} \int_0^{q^*} q f(q) dq$$

Constraints. We solve

$$\begin{aligned} \max_{\alpha, \beta} \quad & U(\alpha, \beta) \\ \text{s.t.} \quad & \alpha_{\min} \leq \alpha \leq \alpha_{\max}, \quad 0 \leq \beta \leq 1, \\ & q^* = \frac{\alpha\mu}{\alpha\mu + (\beta\alpha + 1)(1 - \mu)}, \\ & Q(\alpha, \beta) \geq Q_{\min}, \\ & L(\alpha, \beta) \geq L_{\min} - \xi(\alpha - \alpha_{\min}), \quad \xi \geq 0, \\ & \beta \alpha \bar{m} \bar{p} F(q^*) + \bar{m} \bar{p} L(\alpha, \beta) \leq B_{\text{reward}}, \\ & \alpha \bar{m} \bar{p} F(q^*) \leq B_{\text{penalty}}, \\ & C_{\text{risk}}(\alpha, \beta) \leq C_{\max}. \end{aligned}$$

Solver. We use Sequential Quadratic Programming (SQP) to obtain (α^*, β^*) ; see the example in Appendix D.

D Example Application

We illustrate the optimization framework on real data from the NetEase Youling¹ platform (cf. Section 5.1). We (i) fit worker-ability distributions, (ii) specify a baseline with budgets/constraints, and (iii) solve for (α, β) across scenarios.

Worker Ability Distribution. We estimate the worker ability distribution $f(q)$ using historical data from the platform. As shown in Figure 7, the distribution for different task types can be well-approximated by a Beta distribution: $f(q) = \frac{1}{B(\gamma, \delta)} q^{\gamma-1} (1 - q)^{\delta-1}$, where $B(\gamma, \delta)$ is the Beta function.

Baseline & Scenarios. We define the key parameters and constraints reflecting platform priorities, budget limitations, and task settings. **Baseline configuration.** (a) *Weights:* $w_1 = 0.6$ (quality), $w_2 = 0.5$ (participation), $w_3 = 0.3$ (profitability), $w_4 = 0.2$ (quality-participation interaction), $w_5 = 0.1$ (risk penalty), $w_6 = 0.2$ (penalty factor preference). (b) *Budgets:* $B_{\text{reward}} = 1000$, $B_{\text{penalty}} = 200$. (c) *Constraints:* Minimum quality $Q_{\min} = 0.48$ and minimum participation $L_{\min} = 0.5$. (d) *Worker Ability:* Beta parameters $\gamma = 4$, $\delta = 4$ (average ability $\mu = 0.5$). (e) *Task Prices and Workload:* $1.0 \leq p_j \leq 2.0$ with $E[p_j] = 1.5$; average tasks per worker $\bar{m} = 10$; maximum tasks per worker $m_{\max} = 20$. (f) *Parameter Bounds:* $\alpha \in [0.3, 1.0]$, $\beta \in [0.5, 1]$; risk bound $C_{\max} = 50$, participation factor $\xi = 0.5$. **Scenarios.** We investigate a diverse set of scenarios by fine-tuning parameters within the dynamic staking mode (cf. Eq. (5)). As detailed in Table 4, our analysis covers quality-focused, participation-driven, and budget-restricted environments, as well as scenarios with varying worker abilities and increased task pricing.

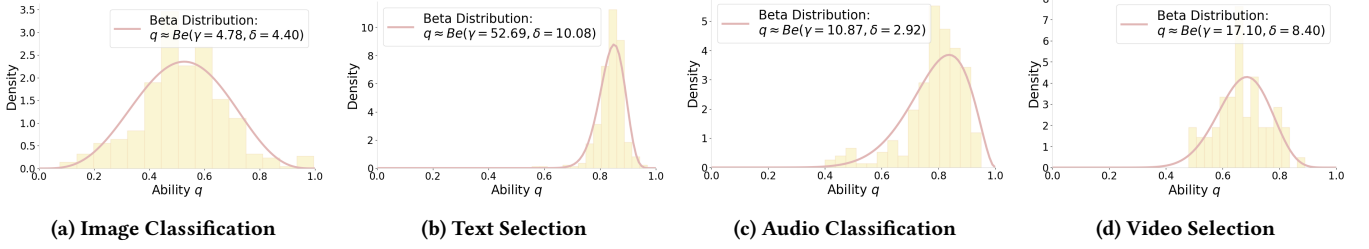


Figure 7: Worker ability distribution across various task types, modeled using Beta distributions.

Table 4: Parameter Adjustments for Different Optimization Scenarios.

Scenario	Description	Parameter Adjustments
Scenario 1: Quality-Focused	The platform places a higher priority on task quality	Increase Task Quality Priority: $w_1 = 0.8$ Decrease Profitability Priority: $w_3 = 0.2$ Keep other weights as in the baseline
Scenario 2: Participation-Focused	The platform emphasizes maximizing worker participation	Lower penalty bound: $\alpha_{\min} = 0.15$ Increase Participation Priority: $w_2 = 0.75$ Relax Quality Floor: $Q_{\min} = 0.45$
Scenario 3: Budget-Constrained	The platform operates under tighter budget constraints	Reduce Reward Budget: $B_{\text{reward}} = 300$ (70% reduction) Reduce Penalty Budget: $B_{\text{penalty}} = 50$ Slightly increase Profitability: $w_3 = 0.35, \beta_{\min} = 0.4$
Scenario 4: Diverse Worker Abilities	Assuming a different worker ability distribution	Adjust Beta distribution parameters: $\gamma = 2, \delta = 5$ Average worker ability decreases to $\mu = \frac{2}{7} \approx 0.2857$ Keep other weights as in the baseline
Scenario 5: Higher Task Prices and Quantity	Tasks are more complex and offer higher rewards	Task Price Range: $2.0 \leq p_j \leq 4.0$ Expected Task Price: $E[p_j] = 3.0$ Increase Average Tasks per Worker: $\bar{m} = 15$

Results & Insights. We solve each scenario via SQP with 30 random restarts. Table 5 reports the optima and metrics; on an 8-core Apple M2 (8 GB), SQP converges in ~ 35 iterations (≈ 4.4 s) per scenario (Scenario 5 is hardest: 76 iters, 9.3 s). **Key takeaways.** (1) *Baseline:* $(\alpha^*, \beta^*) = (0.30, 1.0)$ achieves $Q \approx 0.496$ with high participation $L \approx 0.973$. (2) *Quality priority:* $\alpha^* \uparrow$ to 0.351 slightly reduces L while keeping Q high. (3) *Participation priority:* relaxing $\alpha_{\min}$ yields $\alpha^* = 0.182$, pushing L to 0.992 with $Q \approx 0.499$. (4) *Budget constraint:* tighter budgets drive $\beta^* \downarrow$ to 0.40 and $\alpha^* \uparrow$ to 0.864, trading participation down to $L \approx 0.728$ to remain feasible. (5) *Lower ability:* α^* hits 1.0 (with $\beta^* = 0.50$), preserving feasibility but reducing L to 0.630 and Q to 0.238. (6) *High price/workload:* β^* drops to 0.50 (with $\alpha^* = 0.30$), indicating more penalty retention under higher prices.

E NAI as a Repeated Game: Long-Run Selection

NAI is executed over repeated rounds. In round t , worker i obtains a (possibly negative) net reward U_i^t that is charged to their stake S_i . Let $\mathcal{U}_i(T) = \sum_{t=1}^T U_i^t$ be the cumulative utility. This induces selection: high-quality workers accumulate non-negative utility and persist; persistently low-quality (or collusive) workers accrue losses and exit once $\mathcal{U}_i(T) < -S_i$.

THEOREM E.1 (LONG-RUN SELECTION UNDER NAI). Assume $\{U_i^t\}_{t \geq 1}$ are i.i.d. with finite mean $\mu_i = \mathbb{E}[U_i^t]$.

(1) If $\mu_i \geq 0$, then $\liminf_{T \rightarrow \infty} \mathcal{U}_i(T) \geq 0$ almost surely; worker i retains stake and continues participating.

(2) If $\mu_i = -\delta < 0$, then $\lim_{T \rightarrow \infty} \mathcal{U}_i(T) = -\infty$ almost surely; hence there exists finite T^* with $\mathcal{U}_i(T^*) < -S_i$, forcing exit.

PROOF. By the Strong Law of Large Numbers, $\frac{\mathcal{U}_i(T)}{T} \xrightarrow{\text{a.s.}} \mu_i$. If $\mu_i \geq 0$, cumulative utility does not drift to $-\infty$, giving the limit inferior $\liminf_{T \rightarrow \infty} \mathcal{U}_i(T) \geq 0$. If $\mu_i = -\delta < 0$, then $\mathcal{U}_i(T) \sim -\delta T$ a.s., so some finite T^* satisfies $\mathcal{U}_i(T^*) < -S_i$. $\square$

Implications & Control. NAI induces long-run selection with *bounded downside*: negative payouts apply only to *Staked/Challenging* tasks, while *Beginner* and *Ability* tasks remain non-negative, and any loss is hard-capped by the worker’s stake (opt-in risk). Two knobs regulate the steady state: (i) *Penalty factor* α increases downside and tightens retention toward higher-quality workers; (ii) *Reward coefficient* β redistributes more penalties to high contributors, enlarging the viable set and mitigating deterrence. Operationally, we tune (α, β) via a small 2-variable SQP under platform targets (quality/participation/budget/risk), with safe defaults ($\alpha=0, \beta=1$) and guardrails (participation floor, exposure cap, $\alpha \leq \alpha_{\max}$), solved off the labeling path (Table 5). Negativity is transparent and tied to (ability, correctness), with partial redistribution, yielding accountable incentives without unbounded loss. Robustness to noisy ability estimates is supported by a layered truth-inference stack (historical profiling, gold questions, anchor workers), which dampens noise and collusion in deployment.

Table 5: Optimization Results Under Different Scenarios.

Scenario	α^*	β^*	$Q(\alpha^*, \beta^*)$	$L(\alpha^*, \beta^*)$	$R(\alpha^*, \beta^*)$	$C_{\text{risk}}(\alpha^*, \beta^*)$	$U(\alpha^*, \beta^*)$	Iterations	Runtime (s)	Status
Baseline	0.3000	1.0000	0.4961	0.9733	0.0000	0.0176	0.7762	27	3.227	Feasible
Scenario 1	0.3510	1.0000	0.4940	0.9630	0.0000	0.0313	0.8562	32	4.038	Feasible
Scenario 2	0.1823	1.0000	0.4992	0.9920	0.0000	0.0023	0.9730	37	4.644	Feasible
Scenario 3	0.8637	0.4000	0.4200	0.7275	1.2438	1.0365	0.9360	20	2.470	Feasible
Scenario 4	1.0000	0.5000	0.2381	0.6295	0.7140	0.7140	0.9163	21	2.483	Feasible
Scenario 5	0.3000	0.5000	0.4940	0.9625	0.0544	0.0815	0.7729	76	9.316	Feasible

F Detailed Proofs

Proof of Theorem 4.2

LEMMA F.1 (TRUTHFULNESS). *The NAI mechanism is truthful (incentive-compatible) both in the short term and in the long term.*

PROOF. We prove truthfulness by showing that deviations reduce worker i 's expected net reward.

Short-Term Truthfulness. In the NAI mechanism, each worker's task contribution is evaluated via the AR-CF (Eq. (3)). Rewards, penalties, and net rewards are computed by Eqs. 7–9.

If worker i behaves truthfully, his effective ability is q_i . Deviating by reducing effort or misreporting lowers the effective ability to $q'_i < q_i$. The expected net reward simplifies to the following form (cf. Equation (12)):

$$\mathbb{E}[U_i] \propto \frac{(\beta\alpha + 1)q_i}{\mu} - \frac{\alpha(1 - q_i)}{1 - \mu},$$

which is strictly increasing in q_i . Hence, any deviation reduces expected rewards, establishing short-term truthfulness.

Long-Term Truthfulness. Let $\mathcal{U}_i(T) = \sum_{t=1}^T U_i^t$ represent cumulative utility over T rounds. Any deviation decreases expected per-round reward, leading to:

$$\lim_{T \rightarrow \infty} \frac{\mathcal{U}_i(T)}{T} = \mathbb{E}[U_i^t] < \mathbb{E}[U_i^t]_{\text{truthful}}.$$

Thus, persistent deviations eventually exhaust staked points S_i , confirming long-term truthfulness. $\square$

Other properties of Theorem 4.2—computational efficiency, redistributive incentive and sloppiness—follow directly from the mechanism's design and are therefore omitted for brevity.

Proof of Theorem 4.3

PROOF. For each trusted worker $i \in \mathcal{W}_{\text{trusted}}$, we have $q_i \geq \tau$ and thus assign $w_i = 3q_i \geq 3\tau$. For any colluding worker $j \notin \mathcal{W}_{\text{trusted}}$, $q_j < \tau$ and so $w_j = q_j < \tau$. Define

$$W_{\text{trusted}} = \sum_{i \in \mathcal{W}_{\text{trusted}}} w_i \quad \text{and} \quad W_{\text{colluder}} = \sum_{j \notin \mathcal{W}_{\text{trusted}}} w_j.$$

Then, $W_{\text{trusted}} \geq 3\tau |\mathcal{W}_{\text{trusted}}|$ and $W_{\text{colluder}} < \tau |\mathcal{W}_{\text{colluder}}|$. Taking the ratio, we obtain

$$\frac{W_{\text{trusted}}}{W_{\text{colluder}}} > \frac{3\tau |\mathcal{W}_{\text{trusted}}|}{\tau |\mathcal{W}_{\text{colluder}}|} = 3 \cdot \frac{|\mathcal{W}_{\text{trusted}}|}{|\mathcal{W}_{\text{colluder}}|}.$$

In many practical scenarios, if the number of trusted workers is at least as large as the number of colluders (i.e. $|\mathcal{W}_{\text{trusted}}| \geq |\mathcal{W}_{\text{colluder}}|$), then $\frac{W_{\text{trusted}}}{W_{\text{colluder}}} \geq 3$. Thus, trusted workers dominate the aggregation in the truth inference algorithm, thereby mitigating the influence of colluders. $\square$

Proof of Theorem 4.4

PROOF. Assume that for each gold standard question, a worker i answers correctly with probability s_i . Then, the expected number of correct responses is $\kappa|B|s_i$ and the expected credibility is

$$\mathbb{E}[c_i] = 1 - \frac{1}{\kappa|B|} \cdot (\kappa|B|(1 - s_i)) = s_i.$$

For a colluding worker, suppose $s_i \leq \delta$ with $\delta < 0.7$. By Hoeffding's inequality, for any $\epsilon > 0$,

$$\Pr(c_i \geq \delta + \epsilon) \leq \exp(-2\epsilon^2 \kappa|B|).$$

Setting $\delta + \epsilon = 0.7$ (i.e. $\epsilon = 0.7 - \delta$), the probability that a colluder's credibility c_i exceeds 0.7 is at most $\exp(-2(0.7 - \delta)^2 \kappa|B|)$, which is exponentially small in $\kappa|B|$. Thus, with high probability, colluding workers are filtered out by the threshold $c_{\min} = 0.7$, thereby mitigating collusion. $\square$

Proof of Theorem 4.5

PROOF. For each $k \in \mathcal{A}$, the prior probability is given by

$$P(y_j^* = k) \propto \exp(\lambda n_k),$$

where $n_k = \sum_{i \in \mathcal{W}_{\text{anchor}}} \mathbb{I}(a_{ij} = k)$. Under the assumption that for every $k' \in \mathcal{A} \setminus \{k^*\}$ we have

$$n_{k^*} \geq \left(\sum_{i \in \mathcal{W}_{\text{anchor}}} \mathbb{I}(a_{ij} = k') \right) + 1,$$

it follows that $n_{k^*} - n_{k'} \geq 1$. Thus,

$$\frac{P(y_j^* = k^*)}{P(y_j^* = k')} = \frac{\exp(\lambda n_{k^*})}{\exp(\lambda n_{k'})} = \exp(\lambda(n_{k^*} - n_{k'})) \geq \exp(\lambda).$$

Since $\exp(\lambda) > 1$ for any $\lambda > 0$, the true label k^* is exponentially favored over any other option, thereby mitigating the influence of colluders. $\square$